



Dual-Luciferase® Reporter (DLR) Assay

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Preparation of cell samples

- 1 Culture cells in 24-well cell culture plate until 60% confluency.
- 2 Transfect cells with pRL-TK plasmid (*Renilla* luciferase), ISRE-Luc plasmid (firefly luciferase) and the indicated plasmids, and culture for 24 hours.
- 3 Add IFN- β (1000 U/mL) for another 8 hours.
- 4 Wash cultured cells twice with PBS.
- 5 Lyse cells with 100 μ L 1 $\times$ Passive Lysis Buffer (PLB) from **Dual-Luciferase[®] Reporter Assay System (Promega)**.
- 6 Collect lysates in 1.5 mL centrifuge tubes, rock in 37°C incubator for 15 minutes.
- 7 Centrifuge for 5 minutes at 12,000 g at 4°C.
- 8 Transfer cleared lysates to a new tube prior to reporter enzyme analyses by Dual-Luciferase[®] Reporter Assay System.

Procedure of Dual-Luciferase[®] Reporter (DLR) Assay

- 9 Firefly and *Renilla* luciferase activities were measured by **Dual-Luciferase[®] Reporter Assay System (Promega)**, according to the manufacturer's protocol stated as below.
- 10 Preparation of Luciferase Assay Reagent II
Prepare Luciferase Assay Reagent II (LAR II) by resuspending the provided lyophilized Luciferase Assay Substrate in 10 ml of the supplied Luciferase Assay Buffer II.
- 11 Preparation of Stop & Glo[®] Reagent
Prepare an adequate volume to perform the desired number of DLR[™] Assays (100 μ L reagent per assay). Add 20 μ L of 50X Stop & Glo[®] Substrate to 1ml of Stop & Glo[®] Buffer contained in either a glass vial or siliconized polypropylene tube. Stop & Glo[®] Reagent (Substrate + Buffer) is best when prepared just before use.



- 12 Predisperse 100 μ L of LAR II into the appropriate number of luminometer tubes to complete the desired number of DLR™ Assays.
- 13 Carefully transfer up to 20 μ L of cell lysate into the luminometer tube containing LAR II; mix by pipetting 2 or 3 times. Do not vortex. Place the tube in the luminometer and initiate reading.
- 14 Add 100 μ L of Stop & Glo® Reagent and vortex briefly to mix. Replace the sample in the luminometer, and initiate reading.
- 15 Record the firefly luciferase and *Renilla* luciferase activity measurement.
- 16 Discard the reaction tube, and proceed to the next DLR™ Assay.